

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2165

Hitchcock 138 kV
September 20, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2165. The Interconnection Customer proposes a 46.9 MW battery energy storage facility interconnecting at the Hitchcock 138 kV substation in the Maryland area. The request was received on August 21, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 444 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,549,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Mccall Substaion-Pebbledale 138 KV L
Contingency: Sierra Pacific Lumber 345/1 kV Auto
Loading: 111.4% (Rating: 635 MVA)
DFAX: 5.5%
Pre-project loading: 110.9%
2. Facility: Pebbledale-Taber 138 KV
Contingency: Sierra Pacific Lumber 345/1 kV TX
Loading: 110.3% (Rating: 687 MVA)
DFAX: 5.5%
Pre-project loading: 109.9%
3. Facility: Painters Mill-Hitchcock Substation 138 KV Line
Contingency: Whiteside-Deer Trail 138 KV L
Loading: 106.4% (Rating: 268 MVA)
DFAX: 79.3%
Pre-project loading: 92.5%
4. Facility: Whiteside-Lochern 138 KV
Contingency: Painters Mill-Hitchcock 138 KV
Loading: 104.1% (Rating: 255 MVA)
DFAX: 57.8%
Pre-project loading: 93.4%
5. Facility: Painters Mill-Deer Trail 138 KV L
Contingency: Ramos Substation-Danskammer Generating Sta 138 KV line
Loading: 103.7% (Rating: 219 MVA)
DFAX: 70.1%
Pre-project loading: 88.7%

6. Facility: Deer Trail-Whiteside Substation 138 KV line
 Contingency: Hithcock-Painters Mill 138 KV
 Loading: 102.9% (Rating: 283 MVA)
 DFAX: 57.8%
 Pre-project loading: 93.3%
7. Facility: Northwest-Adamsville 138 KV
 Contingency: Ramos-Danskammer Generating Sta 138 KV
 Loading: 100.9% (Rating: 258 MVA)
 DFAX: 35.7%
 Pre-project loading: 94.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mccall Substaion-Pebbledale 138 KV	Rebuild line with higher-capacity conductor	\$2,367,000	3% (\$75,000)
Pebbledale-Taber 138 KV	Rebuild line with higher-capacity conductor	\$2,851,000	3% (\$90,000)
Painters Mill-Hitchcock Substation 138 KV	Reconductor line	\$1,145,000	87% (\$997,000)
Lochern-Whiteside 138 KV	Reconductor line	\$249,000	81% (\$201,000)
Painters Mill-Deer Trail 138 KV L	Reconductor line	\$177,000	100% (\$177,000)
Deer Trail-Whiteside 138 KV	Reconductor line	\$2,257,000	81% (\$1,824,000)
Northwest-Adamsville 138 KV	Reconductor line	\$1,692,000	70% (\$1,185,000)
	Total allocated to project		\$4,549,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.